University of St.Gallen

Course and Examination Fact Sheet: Spring Semester 2026

8,032: Methods: Social Network Analysis - a Systematic and Quantifying Approach

ECTS credits: 3

Overview examination/s

(binding regulations see below)
decentral - Written work, Digital, Individual work individual grade (50%)
Examination time: Term time
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Examination time: Term time

Attached courses

Timetable -- Language -- Lecturer
8,032,1.00 Methods: Social Network Analysis - a Systematic and Quantifying Approach -- English -- Shahrezaye Morteza

Course information

Prerequisites

This course is assigned to the «Digital Channel & Customer Relationship Management» profile but can also be taken without selecting a specialization.

Basic computer skills are required (no traditional programming skills). This course embraces the "AI Age": we will use LLMs and Web Agents to automate data collection and analysis. Therefore, a willingness to experiment with Generative AI tools and learn basic prompting techniques is a distinct advantage, (maybe) replacing the need for prior mathematical or programming knowledge.

Learning objectives

After successfully participating in the course, students are able to:

- understand the most important theoretical and methodological foundations of social network analysis
- deploy AI Agents and LLMs to gather, clean, and structure real-world data from the web for analysis
- apply these methods to your own research projects using live data experiments rather than static textbook examples
- understand the main concepts of simulation of social systems
- use centrality measures and their calculation to identify key influencers and structural vulnerabilities
- visualize networks and other complex data using industry-standard tools like Gephi

Content

With the advent of social media and the rise of Generative AI, social network analysis (SNA) has become increasingly popular among researchers, journalists, marketers, and the general public. Most of us are familiar with examples of network visualizations such as friendship networks, networks of corrupt politicians, or networks of shell companies.

This course is designed for those who would like to use SNA to view, explore, analyze, and create insights from network data. A key innovation in this iteration of the course is the shift from static, pre-made datasets to dynamic data collection.

Content:

1. Fundamentals of networks
2. Descriptive methods of Social Network Analysis (SNA)

3. AI-Driven Data Acquisition: Scraping the Web with Agents
4. Fundamentals of Gephi
5. Basics of network visualization with Gephi
6. Working with Gephi filters
7. Network statistics
8. Communities and partitioning networks
9. SNA with Cursor (Agentic AI)

Structure and indications of the learning and teaching design

- This course awards 3 ECTS credits, corresponding to a total workload of 90 hours (including self-study, on-campus sessions, and examinations).
- The course is conducted in-person in St. Gallen.

The curriculum is fully self-contained. Comprehensive learning materials, including detailed slides and video tutorials, are provided. The instructor will guide participants through every step of the workflow—from configuring AI agents for data collection to the final analysis in Gephi.

Grading is based on a series of practical exercises and projects. Please note that these are strictly individual tasks (no group work), designed to ensure every student personally masters the use of these digital tools.

Literature

Core Textbooks:

- Yang, S., Keller, F. B., & Zheng, L. (2016). Social network analysis: Methods and examples. Sage Publications.
- Cherven, K. (2015). Mastering Gephi network visualization. Packt Publishing Ltd.

Tutorials on AI Agents & Automated Data Collection:

- **Cursor AI & Coding Agents:**
  - *Cursor AI: A Guide With 10 Practical Examples* – A comprehensive guide on using Cursor's "Agent" mode to write and refactor code via natural language.
  - URL: https://www.datacamp.com/tutorial/cursor-ai-code-editor
- **Anthropic (Claude) Agents:**
  - *Building Agents with the Claude Agent SDK* – Official documentation on how to give Claude "computer use" tools to automate tasks.
  - URL: https://www.anthropic.com/engineering/building-agents-with-the-claude-agent-sdk
- **Google Gemini Agents:**
  - *Google Gemini Enterprise Tutorial: Create AI Agents* – A visual guide on building multi-agent systems and custom orchestrators using Gemini.
  - URL: https://www.youtube.com/watch?v=OOXpW6Aeg_A
- **Graph Construction with LLMs:**
  - *How to Build a Knowledge Graph in Minutes* – A practical tutorial on using LLMs to extract entities and relationships from unstructured text to build network data.
  - URL: https://towardsdatascience.com/enterprise-ready-knowledge-graphs-96028d863e8c/

Additional information

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Examination information

Examination sub part/s

1. Examination sub part (1/2)

Examination modalities

Examination type	Written work
Responsible for organisation	decentral
Examination form	Written work
Examination mode	Digital
Time of examination	Term time
Examination execution	Asynchronous
Examination location	Off Campus
Grading type	Individual work individual grade
Weighting	50%
Duration	--

Languages

Question language: English
Answer language: English

Remark

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Examination-aid rule

Free aids provision

Basically, students are free to choose aids. Any restrictions are defined by the faculty members in charge of the examination under supplementary aids.

Supplementary aids

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2. Examination sub part (2/2)

Examination modalities

Examination type	Written work
Responsible for organisation	decentral
Examination form	Written work
Examination mode	Digital
Time of examination	Term time
Examination execution	Asynchronous
Examination location	Off Campus
Grading type	Individual work individual grade
Weighting	50%
Duration	--

Languages

Question language: English
Answer language: English

Remark
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Examination-aid rule
Free aids provision

Basically, students are free to choose aids. Any restrictions are defined by the faculty members in charge of the examination under supplementary aids.

Supplementary aids
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Content
50% of the final grade is made up of 5 exercises that students must submit during the semester. The other 50% consists of a final project. The students are obliged to work individually.

Basics of network theory and network visualization with Gephi are prerequisites for fulfilling the examination pots. Active participation in the course guarantees student success and does not require them to use material outside of the course.

Literature
The slides provided by the instructor and the presence in the lectures are enough for the examination.

Please note

Please note that only this fact sheet and the examination schedule published at the time of bidding are binding and takes precedence over other information, such as information on StudyNet (Canvas), on lecturers' websites and information in lectures etc.
Any references and links to third-party content within the fact sheet are only of a supplementary, informative nature and lie outside the area of responsibility of the University of St.Gallen.

Documents and materials are only relevant for central examinations if they are available by the end of the lecture period (CW21) at the latest. In the case of centrally organised mid-term examinations, the documents and materials up to CW 13 (Monday, 23rd March 2026) are relevant for testing.

Binding nature of the fact sheets:

- Course information as well as examination date (organised centrally/decentrally) and form of examination: from bidding start in CW 04 (Thursday, 22nd January 2026);
- Examination information (supplementary aids, examination contents, examination literature) for decentralised examinations: in CW 12 (Monday, 16 March 2026);
- Examination information (supplementary aids, examination contents, examination literature) for centrally organised mid-term examinations: in CW 14 (Monday, 30 March 2026);
- Examination information (regulations on aids, examination contents, examination literature) for centrally organised examinations: two weeks before ending with de-registration period in CW 15 (Monday, 06 April 2026).